

Mental Health in Crisis: Changing Patterns in U.S. Emergency Departments

Research Report

Exploratory Data Analysis of CDC National Syndromic Surveillance Program Data in Mental Health-Related ED Visit Rates from 2022-2026.

Introduction

Emergency departments play a critical role in the U.S. healthcare system, but increasing patient demand and limited capacity can create challenges involving crowding, prolonged wait times, and access to timely care. Mental health-related visits are particularly important within this environment because patients may require specialized evaluation, treatment, or placement that is not always immediately available. Historically, mental health conditions have represented a growing component of emergency care utilization, increasing the importance of understanding how behavioral health demand is changing within emergency departments.

However, changes in mental health-related ED utilization may not occur equally across all conditions. Anxiety, depression, trauma-related disorders, bipolar disorders, schizophrenia spectrum disorders, and suspected suicide attempts may each follow different patterns over time. Examining these conditions individually can provide a more detailed view of where mental health-related ED demand may be increasing, decreasing, or fluctuating and can help identify areas that warrant closer monitoring.

This report analyzes monthly U.S. emergency department visit rates for six major mental health conditions using data from the CDC National Syndromic Surveillance Program (NSSP). The analysis focuses on January 2022 through May 2026 and evaluates both changes between the beginning and end of the study period and trends across monthly observations. The goal is to identify which mental health conditions experienced the largest changes in ED visit rates and determine whether those changes reflect consistent patterns over time.

Business and Public Health Relevance

Understanding changes in mental health-related emergency department utilization is important for both healthcare operations and public health planning. Mental health conditions already represent a substantial component of emergency care demand in the United States. Weiss et al. (2020) estimated that approximately 8.4 million of 100.9 million annual U.S. emergency department visits between 2007 and 2016 involved psychiatric or substance-use diagnoses. The proportion of ED visits involving these diagnoses increased from 6.6% to 10.9% over the study period, highlighting the growing role of behavioral health within emergency care.

The operational impact of mental health-related ED utilization extends beyond visit volume. Mental health patients may require extended evaluation, treatment, or placement while appropriate services become available. Hoffmann et al. (2025) estimated 5.9 million pediatric mental health ED visits nationally from 2018 through 2022. Among visits resulting in admission or transfer, 32.1% involved ED stays of at least

12 hours. Prolonged stays such as these can occupy ED capacity and resources for extended periods and contribute to operational challenges associated with mental health boarding.

Access to appropriate emergency care is another concern. Sangal et al. (2025), analyzing 148.6 million U.S. ED encounters, found that emergency care access failures were higher among patients with mental health conditions. Pediatric patients with mental health diagnoses experienced a 10.5-percentage-point increase in access failures compared with a 6.2-percentage-point increase among pediatric patients without mental health diagnoses. The study also identified mental health bed capacity as an important structural issue affecting emergency care access.

Against this broader healthcare context, identifying which mental health conditions are changing most may help healthcare organizations recognize areas where demand is evolving. Condition-specific patterns can provide information relevant to staffing, resource allocation, behavioral health services, psychiatric capacity, and operational planning. From a public health perspective, these trends may also help identify areas that warrant additional surveillance, prevention efforts, outpatient services, or community-based mental health resources.

The objective of this analysis does not establish that changes in individual mental health conditions cause ED crowding or access problems. Instead, it provides a data-driven view of how the composition of mental health-related ED utilization has changed over time. Combined with existing evidence of mental health-related capacity and access challenges, these findings can help identify conditions that may warrant closer monitoring and further investigation.

Research Question

How did emergency department visit rates for major mental health conditions change between January 2022 and May 2026, and which conditions exhibited the largest observed changes?

Executive Summary

This exploratory analysis examines monthly U.S. emergency department (ED) visit rates for six mental health conditions using National Syndromic Surveillance Program data. The analysis focuses on overall national trends and supplements the primary analysis with descriptive summaries by age, sex, and race/ethnicity. Endpoint growth from January 2022 to May 2026 was highest for Trauma and Stressor-related Disorders (+15.3%), followed by Anxiety Disorders (+5.7%) and Suspected Suicide Attempts (+3.2%). However, simple linear regression did not find statistically significant upward linear trends for Trauma and Stressor-related Disorders or Anxiety Disorders. Depressive, Bipolar, and Schizophrenia Spectrum Disorders showed statistically significant downward linear trends over the study period. These results suggest that endpoint growth and long-run linear trend can tell different stories and should be interpreted together.

1. Data Source and Study Scope

The analysis uses the CDC National Syndromic Surveillance Program (NSSP) Mental Health-Related Emergency Department Visit Rates dataset. The raw file contained 10,591 rows and five variables: month_end, condition, demographics_type, demographics_values, and rate_per_100000_visits. The rate variable represents mental health-related ED visits per 100,000 ED visits. The notebook contains observations extending beyond the focal period, but the primary trend and growth analyses were restricted to January 2022 through May 2026.

Six individual conditions were retained for condition-level analysis: Anxiety Disorders, Bipolar Disorders, Depressive Disorders, Schizophrenia Spectrum Disorders, Suspected Suicide Attempts, and Trauma and Stressor-related Disorders. The aggregate “Any Mental Health” category was excluded because it combines mental health-related visits rather than representing one specific condition.

2. Data Cleaning and Preparation

Initial inspection found no missing values. The month_end field was converted from year-month text to a datetime variable. Potential outliers were first evaluated using the interquartile range (IQR) method. Because the dataset combines age, sex, race/ethnicity, and total-population observations, many initially flagged values reflected genuine differences among demographic groups rather than obvious data-entry errors. The primary national analysis therefore used only demographics_type = “Total.”

For supporting demographic analyses, the cleaned data were separated into age, sex, and race/ethnicity subsets. Aggregate “All” rows were removed from these demographic subsets so that comparisons represented distinct groups rather than mixing group-specific and aggregate observations.

3. Descriptive Findings

Across the overall national condition data, Anxiety Disorders had the highest mean ED visit rate (2,713.99 per 100,000 visits) and the greatest standard deviation (362.08), indicating the largest average rate and substantial month-to-month variability among the six conditions. Suspected Suicide Attempts had the lowest mean rate (151.76 per 100,000 visits).

Condition	Mean rate	Std. dev.	N
Anxiety Disorders	2,713.99	362.08	89
Depressive Disorders	1,821.67	324.08	89
Bipolar Disorders	540.61	111.68	89
Schizophrenia Spectrum Disorders	450.29	71.06	89
Trauma and Stressor-related Disorders	419.84	42.70	89
Suspected Suicide Attempts	151.76	16.37	89

Supporting Demographic Patterns

When all six conditions and available months were summarized within demographic groups, the 12–17 age group had the highest mean rate (1,237.12), while adults age 65+ had the lowest (933.83). Females had a higher mean rate than males (1,122.98 versus 896.41). Among race/ethnicity categories, White observations had the highest mean rate (1,226.52), followed by AI/AN observations (1,191.92), while Asian observations had the lowest mean rate (608.25). These demographic summaries pool multiple conditions, so they should be treated as broad descriptive context rather than condition-specific causal comparisons.

4. Trend and Growth Analysis

The monthly trend analysis tracks each condition from January 2022 through May 2026 using the overall U.S. population subset. Visual inspection shows substantial month-to-month variation, with different conditions following different trajectories over time.

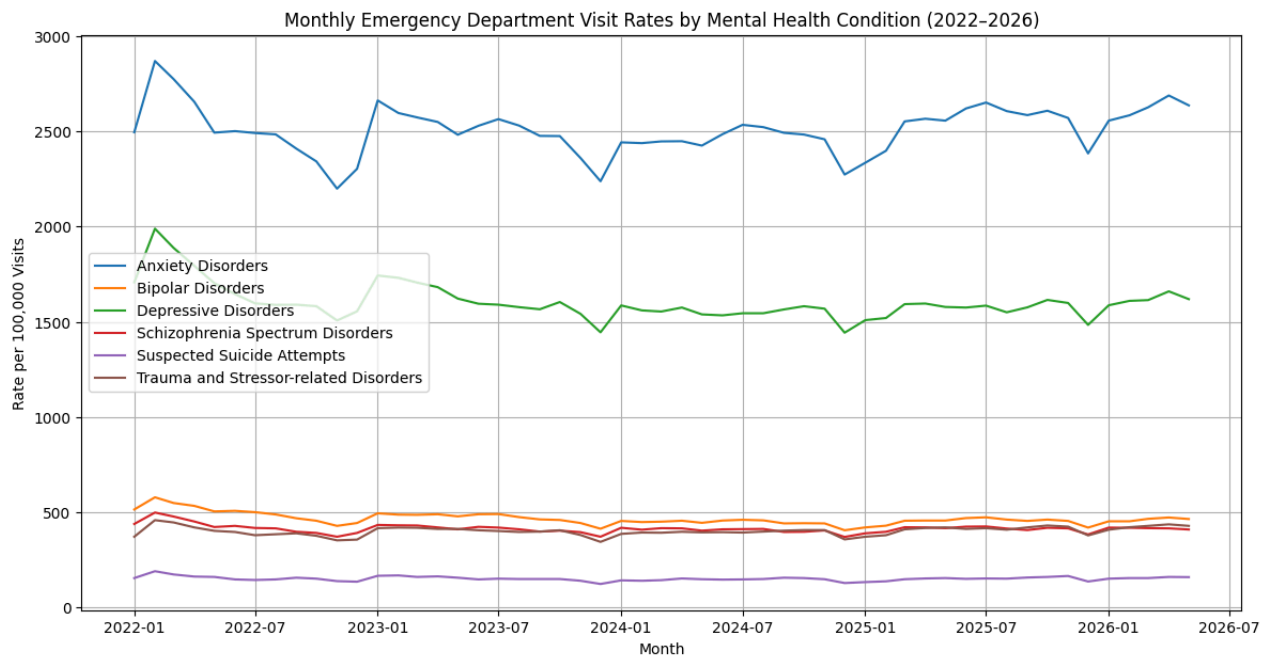


Figure 1. Monthly emergency department visit rates by mental health condition, January 2022–May 2026. Rates are shown per 100,000 ED visits for the overall U.S. population.

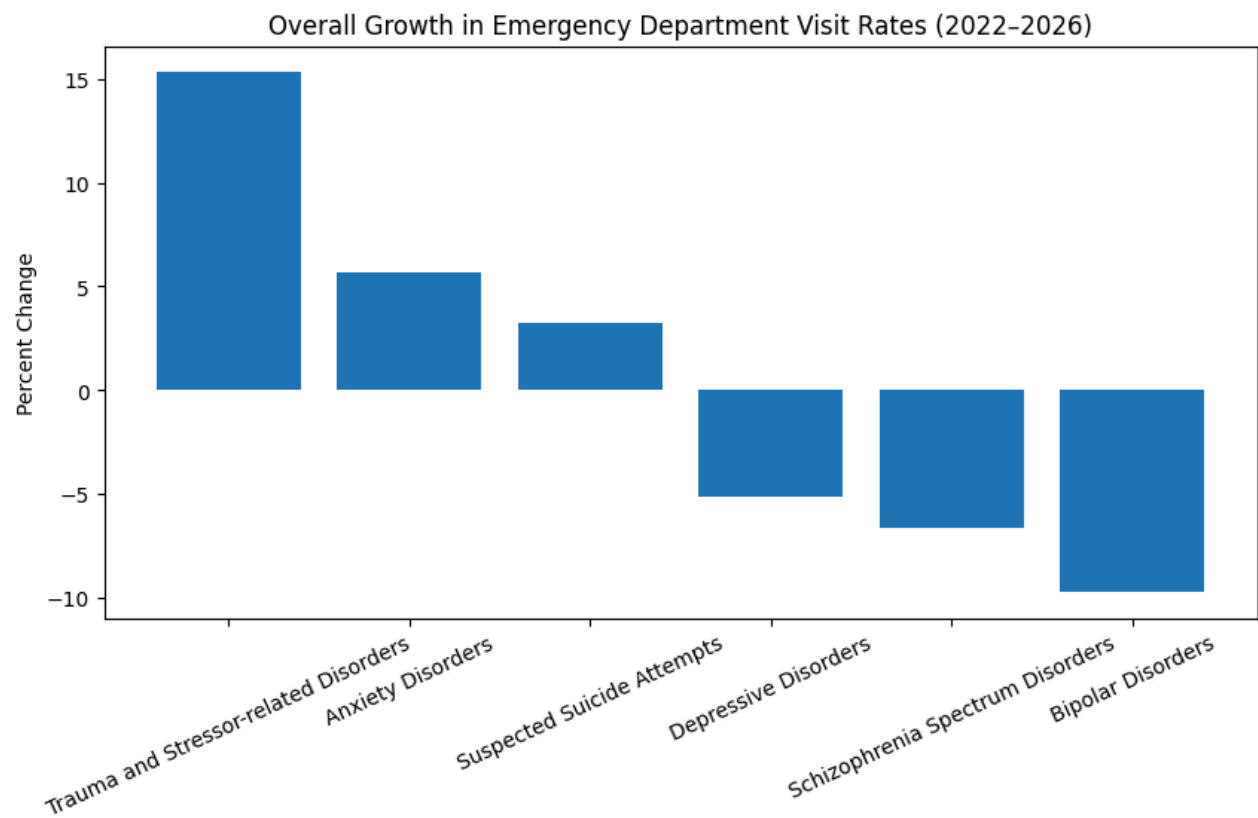


Figure 2. Percent change in emergency department visit rates by condition from January 2022 to May 2026. Positive values indicate endpoint growth; negative values indicate endpoint decline.

Condition	Absolute change	Percent change
Trauma and Stressor-related Disorders	+57	+15.3%
Anxiety Disorders	+141	+5.7%
Suspected Suicide Attempts	+5	+3.2%
Depressive Disorders	-88	-5.2%
Schizophrenia Spectrum Disorders	-29	-6.6%
Bipolar Disorders	-50	-9.7%

Table 2. Endpoint change from January 2022 to May 2026.

Trauma and Stressor-related Disorders had the largest relative endpoint increase at approximately 15.3%. Anxiety Disorders increased by approximately 5.7%, while Suspected Suicide Attempts increased by approximately 3.2%. Bipolar Disorders had the largest relative decline at approximately 9.7%. Because May 2026 is the last available month, these values compare endpoints rather than complete calendar years.

5. Inferential Analysis: Linear Regression

To test whether the full set of monthly observations supported a consistent linear trend, a separate simple linear regression was fit for each condition. Month number was used as the independent variable and rate_per_100000_visits as the dependent variable. The null hypothesis was that the slope equals zero (no linear relationship between time and visit rate); the alternative hypothesis was that the slope differs from zero. Statistical significance was evaluated at $\alpha = 0.05$.

Condition	Slope	R ²	p-value	Decision ($\alpha=.05$)
Anxiety Disorders	0.7636	0.0088	0.5034	Fail to reject H ₀
Trauma and Stressor-related Disorders	0.3287	0.0482	0.1140	Fail to reject H ₀
Suspected Suicide Attempts	-0.1280	0.0309	0.2080	Fail to reject H ₀
Schizophrenia Spectrum Disorders	-0.4502	0.0995	0.0214	Reject H ₀
Bipolar Disorders	-1.2209	0.3440	<0.001	Reject H ₀
Depressive Disorders	-2.8966	0.2192	0.0004	Reject H ₀

Table 3. Simple linear regression results for monthly ED visit rates, January 2022–May 2026.

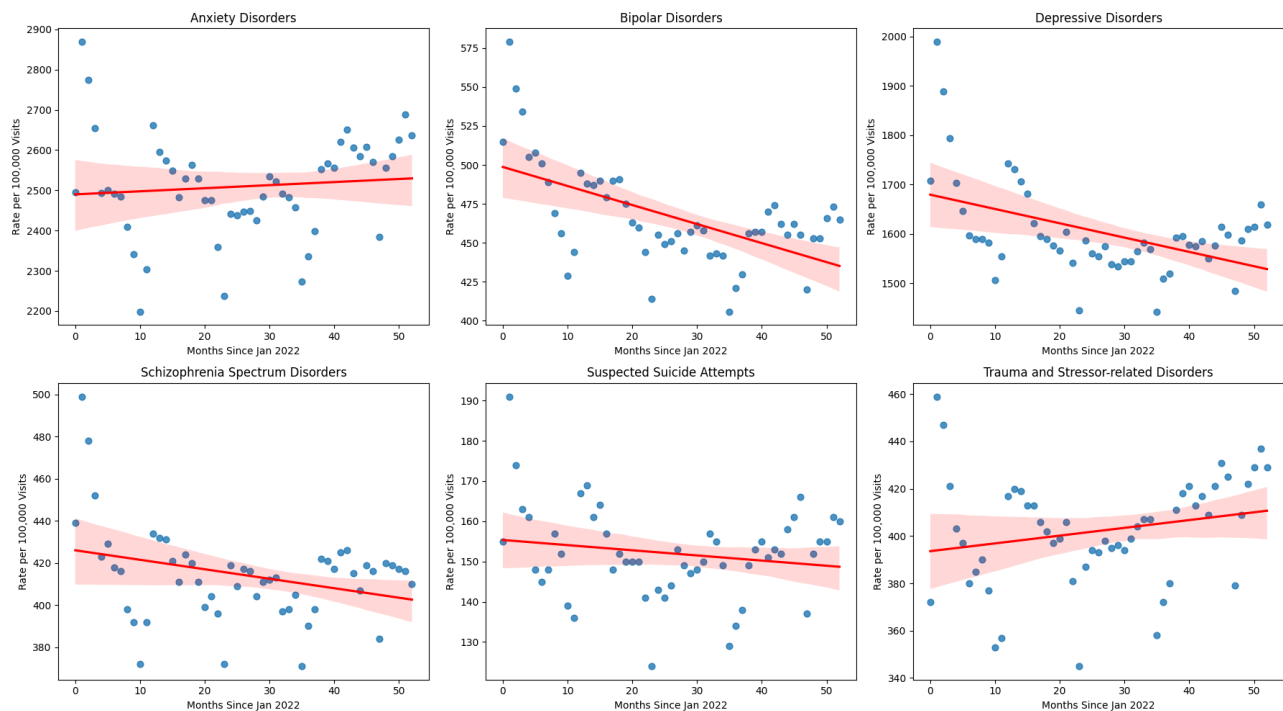


Figure 3. Condition-specific simple linear regressions of monthly ED visit rates against months since January 2022. The fitted line summarizes the estimated linear direction over the study period.

Anxiety Disorders and Trauma and Stressor-related Disorders had positive estimated slopes, but their p-values were greater than 0.05, so the analysis did not provide sufficient evidence of a consistent upward linear trend. Suspected Suicide Attempts also did not show a statistically significant linear trend. Schizophrenia Spectrum, Bipolar, and Depressive Disorders had negative slopes with p-values below 0.05, providing evidence of downward linear trends over the observed period. Depressive Disorders had the steepest estimated decline (-2.8966 rate points per month).

6. Discussion

The most important finding is the difference between endpoint growth and regression-based trend evidence. Trauma and Stressor-related Disorders showed the largest increase when comparing January 2022 with May 2026, yet the regression result was not statistically significant. Anxiety Disorders showed a similar pattern: a higher endpoint in May 2026 than in January 2022, but little evidence that the entire monthly series followed a steady linear increase. This indicates that the beginning-to-end change may be influenced by fluctuations within the series and should not be interpreted as proof of a sustained upward trend.

In contrast, Depressive, Bipolar, and Schizophrenia Spectrum Disorders produced statistically significant negative slopes. For planning purposes, the analysis therefore distinguishes conditions that ended the period above their January 2022 level from conditions that demonstrated a more consistent directional pattern across all monthly observations.

7. Limitations

- 2026 is incomplete: the dataset contains observations only through May 2026, so the endpoint analysis compares January 2022 with May 2026 rather than two complete years.

- The growth calculation is sensitive to the selected start and end months and does not summarize all variation occurring between those endpoints.
- Simple linear regression captures only a linear relationship with time. Seasonal patterns, abrupt changes, nonlinear trends, and autocorrelation in monthly observations are not modeled.
- Demographic descriptive statistics pool multiple mental health conditions. They provide context but do not identify which specific condition drives differences between demographic groups.
- The analysis is observational and descriptive; it does not establish causes of changes in ED visit rates.

8. Conclusion and Future Scope

From January 2022 through May 2026, Trauma and Stressor-related Disorders had the largest relative endpoint increase in national mental health-related ED visit rates (+15.3%), followed by Anxiety Disorders (+5.7%). These increases were not accompanied by statistically significant positive linear trends, suggesting that continued monitoring is more appropriate than concluding that a steady upward trend has already been established. Depressive, Bipolar, and Schizophrenia Spectrum Disorders showed statistically significant downward linear trends during the study period.

Future analysis could extend the time series as additional 2026 data become available, evaluate seasonality and nonlinear trends, and examine condition-specific patterns within age, sex, and race/ethnicity groups. Those analyses would help determine whether the endpoint increases observed for trauma- and anxiety-related visits persist and whether particular populations are driving the changes.

References

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